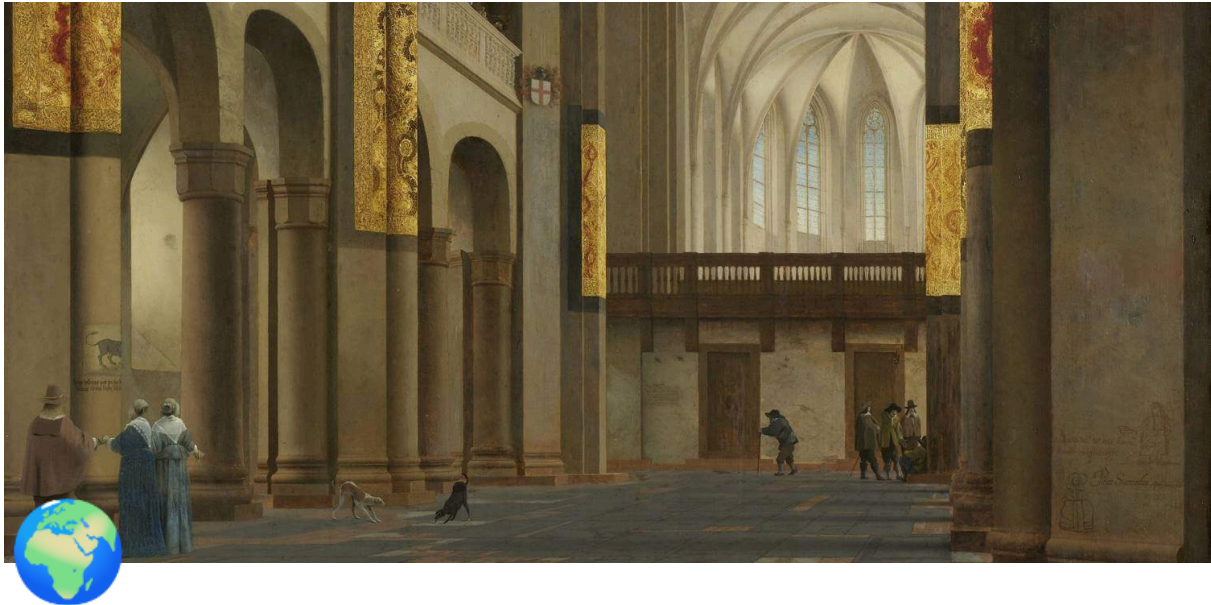




NISC

Tutorial-11,12,13

Tutorial - 11

Conceptual Questions

A. What is the main idea behind Ant Colony Optimization?

ACO is a probabilistic metaheuristic inspired by how real ants find shortest paths using pheromone trails. Artificial ants build solutions iteratively, and better solutions reinforce pheromone on their paths, guiding future ants toward optimal solutions.

B. What is pheromone in ACO and why is it important?

Pheromone is a numerical value on each path representing its attractiveness based on past solutions. It acts as indirect communication (stigmergy) between ants — higher pheromone means more ants will prefer that path, enabling collective learning.

C. Why is ACO useful for problems like TSP?

TSP is a combinatorial optimization problem where exhaustive search is infeasible for large inputs. ACO explores multiple routes simultaneously through

its ant population and converges toward near-optimal tours using pheromone-guided probabilistic decisions.

D. Why do pheromones evaporate in ACO?

Evaporation prevents the algorithm from converging too early on a suboptimal solution (stagnation). It gradually reduces pheromone on unused paths, allowing exploration of new routes and forgetting poor past decisions.

E. How do ants choose between two possible paths in ACO?

Each ant uses a probabilistic formula based on pheromone level (τ) raised to α and visibility (η) raised to β . The path with higher combined score gets a higher selection probability, but randomness still allows exploration of less-traveled paths.

Numerical Questions

A. Path Selection Probability

Given:

- Path A: $\tau_A = 3.0$, $\eta_A = 0.5$
- Path B: $\tau_B = 1.5$, $\eta_B = 1.0$
- $\alpha = 1$, $\beta = 1$

Formula:

$$P_i = \frac{\tau_i^\alpha \cdot \eta_i^\beta}{\sum_j \tau_j^\alpha \cdot \eta_j^\beta}$$

Step 1: Compute numerator for Path A

$$\tau_A^\alpha \cdot \eta_A^\beta = 3.0^1 \times 0.5^1 = 3.0 \times 0.5 = 1.5$$

Step 2: Compute numerator for Path B

$$\tau_B^\alpha \cdot \eta_B^\beta = 1.5^1 \times 1.0^1 = 1.5 \times 1.0 = 1.5$$

Step 3: Compute the denominator (sum of both)

$$\sum_j \tau_j^\alpha \cdot \eta_j^\beta = 1.5 + 1.5 = 3.0$$

Step 4: Compute P_A

$$P_A = \frac{1.5}{3.0} = 0.5$$

Step 5: Compute P_B

$$P_B = \frac{1.5}{3.0} = 0.5$$

Result: P_A = 0.5 (50%), P_B = 0.5 (50%)

Both paths are equally likely to be chosen.

B. Pheromone Update After Evaporation and Deposit

Given:

- $\tau_{old} = 2.0$
- $\rho = 0.5$
- $L_1 = 8, L_2 = 4$
- $Q = 20$

Formula:

$$\tau^{new} = (1 - \rho) \cdot \tau^{old} + \sum_k \Delta\tau^k$$

$$\text{where } \Delta\tau^k = \frac{Q}{L_k}$$

Step 1: Apply evaporation

$$(1 - \rho) \cdot \tau^{old} = (1 - 0.5) \times 2.0 = 0.5 \times 2.0 = 1.0$$

Step 2: Compute pheromone deposit by Ant 1 ($L_1 = 8$)

$$\Delta\tau^1 = \frac{Q}{L_1} = \frac{20}{8} = 2.5$$

Step 3: Compute pheromone deposit by Ant 2 ($L_2 = 4$)

$$\Delta\tau^2 = \frac{Q}{L_2} = \frac{20}{4} = 5.0$$

Step 4: Compute total deposit

$$\sum_k \Delta\tau^k = 2.5 + 5.0 = 7.5$$

Step 5: Compute τ_{new}

$$\tau^{new} = 1.0 + 7.5 = 8.5$$

Result: $\tau_{new} = 8.5$

Tutorial - 12

Conceptual Questions

A. What is the main idea of Particle Swarm Optimization (PSO)?

PSO is a population-based metaheuristic inspired by the social behavior of birds flocking or fish schooling. Each particle moves through the search space by updating its velocity based on its own best experience and the swarm's best experience, collectively converging toward the optimal solution.

B. What are pbest and gbest in PSO?

pbest (personal best) is the best position a particle has visited in its own history. gbest (global best) is the best position found by any particle in the entire swarm. Both act as attractors that guide each particle's movement toward better regions of the search space.

C. Explain how inertia weight (w) balances exploration and exploitation.

A high w keeps the particle moving fast with little change in direction, encouraging exploration of new regions. A low w slows the particle down, allowing it to fine-tune around known good positions (exploitation). Typically w is decreased over iterations to shift from exploration early on to exploitation later.

D. Why might PSO converge faster than random search?

PSO uses structured information sharing — particles remember their own best positions and communicate the global best. This biases movement toward promising regions instead of searching blindly, making convergence significantly faster than random search which has no memory or cooperation.

Numerical Question

A. Single-Dimension Particle Update

Given:

- $x = 2.0, v = 0.5$
- $pbest = 1.5, gbest = 1.0$
- $w = 0.7, c_1 = 1.5, c_2 = 1.5$
- $r_1 = 0.6, r_2 = 0.3$

Formula:

$$v_{new} = w \cdot v + c_1 r_1 (pbest - x) + c_2 r_2 (gbest - x)$$

$$x_{new} = x + v_{new}$$

Step 1: Compute the inertia term

$$w \cdot v = 0.7 \times 0.5 = 0.35$$

Step 2: Compute the cognitive term (attraction to pbest)

$$\begin{aligned} c_1 \cdot r_1 \cdot (pbest - x) &= 1.5 \times 0.6 \times (1.5 - 2.0) \\ &= 1.5 \times 0.6 \times (-0.5) \\ &= 0.9 \times (-0.5) = -0.45 \end{aligned}$$

Step 3: Compute the social term (attraction to gbest)

$$\begin{aligned} c_2 \cdot r_2 \cdot (gbest - x) &= 1.5 \times 0.3 \times (1.0 - 2.0) \\ &= 1.5 \times 0.3 \times (-1.0) \\ &= 0.45 \times (-1.0) = -0.45 \end{aligned}$$

Step 4: Compute v_new

$$\begin{aligned} v_{new} &= 0.35 + (-0.45) + (-0.45) \\ v_{new} &= 0.35 - 0.45 - 0.45 \end{aligned}$$

$v_{new} = -0.55$

Step 5: Compute x_new

$$x_{new} = x + v_{new} = 2.0 + (-0.55)$$

$x_{new} = 1.45$

Interpretation: The particle was at $x = 2.0$, which is to the right of both pbest (1.5) and gbest (1.0). So both cognitive and social forces pulled it left (negative direction), resulting in a negative velocity and the particle moving left to $x = 1.45$ — closer to the optimal region.

Tutorial - 13

Conceptual Questions

A. How bees search and how it maps to optimization:

Real bees scout for food, return to the hive, and share information through waggle dances — the better the source, the more bees it attracts. In optimization, food sources = candidate solutions, nectar amount = fitness

value, and bees collectively converge toward the best solution through guided search and information sharing.

B. Three bee types in ABC:

Employed bees are assigned to specific food sources and search their neighborhood for better solutions. Onlooker bees wait at the hive and probabilistically choose a food source based on nectar (fitness) shared by employed bees. Scout bees replace abandoned food sources by randomly generating new solutions when a source hasn't improved after a set number of trials (limit).

C. How an employed bee generates a new candidate:

It picks a random neighbor solution k and a random dimension j , then perturbs the current solution using: $v_{ij} = x_{ij} + \varphi(x_{ij} - x_{kj})$, where φ is a random number in $[-1, 1]$. If the new candidate is better, it replaces the old one (greedy selection).

D. How onlooker bees choose a food source:

Onlookers select food sources using roulette-wheel (fitness-proportionate) selection — sources with higher nectar get higher probability. This biases the swarm toward exploiting better-known regions while still allowing weaker sources a small chance of selection.

E. The "limit" concept and scout replacement:

Each food source tracks how many consecutive trials it has gone without improvement. If this count exceeds a threshold called "limit," the employed bee abandons the source and becomes a scout, randomly generating a brand new solution. This prevents stagnation on local optima.

F. Why ABC is population-based and how it balances exploration/exploitation:

ABC maintains a population of food sources (solutions) updated simultaneously, making it population-based. Employed and onlooker bees exploit known good sources through neighborhood search, while scout bees explore entirely new random regions — the limit parameter controls how long exploitation is attempted before switching to exploration.

Numerical Questions

A. Onlooker Selection Probabilities

Given: nectar = {10, 25, 15}

Formula:

$$p_i = \frac{nectar_i}{\sum_j nectar_j}$$

Step 1: Compute total nectar

$$\sum nectar = 10 + 25 + 15 = 50$$

Step 2: Compute p1

$$p_1 = \frac{10}{50} = 0.20$$

Step 3: Compute p2

$$p_2 = \frac{25}{50} = 0.50$$

Step 4: Compute p3

$$p_3 = \frac{15}{50} = 0.30$$

Verification: $0.20 + 0.50 + 0.30 = 1.0 \checkmark$

Result: $p_1 = 0.20, p_2 = 0.50, p_3 = 0.30$

B. Employed Bee Neighborhood Move & Acceptance

Given:

- $x_i = (2.0, 3.5), x_k = (1.0, 4.0)$
- $\varphi_1 = -0.4, \varphi_2 = 0.2$
- $f(x) = (x_1 - 1)^2 + (x_2 - 2)^2$

Formula (dimension-wise):

$$v_{ij} = x_{ij} + \phi_j(x_{ij} - x_{kj})$$

Step 1: Compute v for dimension 1 (j=1)

$$\begin{aligned} v_{i1} &= x_{i1} + \phi_1(x_{i1} - x_{k1}) \\ &= 2.0 + (-0.4)(2.0 - 1.0) \\ &= 2.0 + (-0.4)(1.0) \\ &= 2.0 - 0.4 = 1.6 \end{aligned}$$

Step 2: Compute v for dimension 2 (j=2)

$$\begin{aligned} v_{i2} &= x_{i2} + \phi_2(x_{i2} - x_{k2}) \\ &= 3.5 + (0.2)(3.5 - 4.0) \\ &= 3.5 + (0.2)(-0.5) \end{aligned}$$

$$= 3.5 - 0.1 = 3.4$$

So the candidate solution is: $v_i = (1.6, 3.4)$

Step 3: Evaluate $f(x_i)$ — current solution

$$\begin{aligned} f(x_i) &= (2.0 - 1)^2 + (3.5 - 2)^2 \\ &= (1.0)^2 + (1.5)^2 \\ &= 1.0 + 2.25 = 3.25 \end{aligned}$$

Step 4: Evaluate $f(v_i)$ — candidate solution

$$\begin{aligned} f(v_i) &= (1.6 - 1)^2 + (3.4 - 2)^2 \\ &= (0.6)^2 + (1.4)^2 \\ &= 0.36 + 1.96 = 2.32 \end{aligned}$$

Step 5: Acceptance Check

Since lower is better:

$$f(v_i) = 2.32 < f(x_i) = 3.25$$

The condition $f(v_i) < f(x_i)$ is TRUE → The new candidate $v_i = (1.6, 3.4)$ is ACCEPTED and replaces x_i .

Interpretation: The employed bee successfully found a better food source closer to the global optimum (1, 2), so it moves to the new position.